

ADC Peptide Mapper

User Manual — Version 0.8

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Contents

1	Quick Start	4
2	What's New in v0.8	5
3	Installation and Tool Acquisition	5
3.1	R and Shiny Packages	5
3.2	MS Amanda 3.0 (Primary Search Engine)	6
3.3	Tide / Crux 4.x (Fallback Search Engine)	7
3.4	ProteoWizard MSConvert (Raw File Conversion)	7
3.5	Anthropic API Key (AI Assistant)	8
3.6	Background Proteome Database	8
3.7	Summary Table	8
4	Application Overview	9
5	Tab 1 — Input and Setup	9
5.1	ADC FASTA Input	9
5.2	ADC Name	10
5.3	Enzyme and Cleavage Parameters	10
5.4	Background Proteome	10
5.5	Run Digest	10
6	Tab 2 — Modifications	11
6.1	Fixed Modifications	11
6.2	Variable Modifications	11
6.3	Linker Biotransformation Modifications (new in v0.8)	11
6.4	Special Modifications	11
6.5	Custom Modification Builder	12
6.6	DAR Distribution Settings (new in v0.8)	12
7	Tab 3 — Peptide Results	12
7.1	Value Boxes	12
7.2	Peptide Results Table	12
7.3	Filters	12

7.4	Cross-Species Co-Uniqueness Check (new in v0.8)	12
7.5	Excel Export	13
7.6	Sequence Coverage Map	13
8	Tab 4 — Transition List	14
8.1	Instrument Platform	14
8.2	DAR Level Selection (new in v0.8)	14
8.3	Ion Parameters	14
8.4	Averagine Isotope Envelope (new in v0.8)	14
8.5	Filters and Options	14
8.6	Downloads	15
9	Tab 5 — Heavy Labelling	15
9.1	Isotope Label Presets	15
9.2	Charge States	15
9.3	Output	15
9.4	Downloads	15
10	Tab 6 — MS/MS Search	16
10.1	Engine Status Badge	16
10.2	Engine Executable Path	16
10.3	Spectral File Upload	16
10.4	Score Filters	17
10.5	Target-Decoy FDR Estimation (new in v0.8)	17
10.6	MS Amanda Settings	17
10.7	Results	18
10.8	Downloads	18
11	Tab 7 — AI Assistant (new in v0.8)	18
11.1	Setup	18
11.2	Context Toggle	19
11.3	API Key Security	19
12	Output Reference	19
12.1	Peptide Table Columns	19
12.2	Transition List Columns	19
13	MS/MS Search Engine Setup	20
13.1	MS Amanda 3.0 — Windows	20
13.2	MS Amanda 3.0 — Linux	20
13.3	MS Amanda 3.0 — macOS	21
13.4	Tide / Crux — All Platforms	21
13.5	Setting Environment Variables in R	21
14	Deployment to shinyapps.io	22
15	Running Unit Tests and Mass Accuracy Benchmark	22
16	Scientific Background	23

16.1	Monoisotopic Masses	23
16.2	Peptide Mass Calculation	23
16.3	Fragment Ion Calculation	23
16.4	Average Isotope Envelope	23
16.5	Uniqueness Checking	24
16.6	Target-Decoy FDR Estimation	24
16.7	MS Amanda Scoring	24
16.8	Tide XCorr Scoring	24
17	Troubleshooting	24
17.1	App does not launch	24
17.2	Background database build fails	24
17.3	Digest error: “the condition has length > 1”	25
17.4	AI Assistant returns API error	25
17.5	MS Amanda not detected	25
17.6	Mass accuracy is off by ~1 Da	25
17.7	shinyapps.io deployment bundle too large	25
18	Citation	25

ADC Peptide Mapper is an open-source R Shiny application for in-silico proteolytic digest, peptide uniqueness checking, Drug-to-Antibody Ratio (DAR) distribution modelling, linker biotransformation variable modifications, instrument-specific transition list generation, heavy isotope labelling, MS/MS search confirmation, and AI-assisted interpretation for Antibody-Drug Conjugates (ADCs).

This manual covers installation, configuration, and operation of all seven tabs, including the new DAR modelling module (Tab 2 / Tab 4), linker biotransformation modifications (Tab 2), the AI Assistant (Tab 7), and the updated MS/MS search module (Tab 6) with MS Amanda 3.0 as the primary engine and Tide/Crux as the automatic fallback.

1 Quick Start

The following steps get the app running with the built-in Trastuzumab demo sequence in under five minutes, assuming R is already installed.

Step 1. Install R packages (one-time):

```
install.packages(c(
  'shiny', 'bs4Dash', 'DT', 'data.table', 'openxlsx',
  'httr2', 'stringr', 'dplyr', 'shinycssloaders',
  'shinyjs', 'htmltools'
))
install.packages('XML')
```

Step 2. Build background databases (one-time, ~10 min, requires internet):

```
setwd('path/to/ADC_Peptide_Mapper_v0.8')
source('build_background_db.R')
```

Step 3. Launch the app:

```
shiny::runApp('app.R')
```

Step 4. The demo Trastuzumab HC + LC sequence is pre-loaded. Click **Run Digest & Uniqueness Check** (bottom of the FASTA Input card) to generate peptides.

Step 5. Navigate to Tab 3 to view the peptide table and the Sequence Coverage Map.

Step 6. Go to Tab 4 to select an instrument platform and download the transition list.

Step 7. (Optional) Go to Tab 7, enter your Anthropic API key, and use the AI Assistant for interpretation and method guidance.

2 What's New in v0.8

The following changes were made relative to v0.7:

Feature	v0.7	v0.8
Tabs	6	7 (+ AI Assistant)
DAR modelling	—	DAR0–DARn with per-level MRM transition lists
Conjugation chemistry	—	Cysteine thiol-maleimide, lysine NHS-ester / hydrazone, site-specific
Linker biotransformations	—	5 variable mods (see Section 6.3)
Isotope envelope	—	Averagine-based (Senko 1995)
FDR estimation	—	Target-decoy FDR (Kall 2008)
Cross-species filtering	—	Co-species uniqueness (Human / Cyno / Rat)
cRAP contaminants	—	Integrated into background build
MS/MS search engine	MSFragger (commercial)	MS Amanda 3.0 (free) + Tide/Crux (fallback)
Mass accuracy benchmark	—	Sub-mDa vs IgG1 tryptic reference peptides
Unit tests	—	79 tests, 100% pass rate
AI Assistant	—	Tab 7: Claude API chat with ADC context
Sidebar citation	—	Author, DOI, contact always visible
Run Digest placement	Separate row	Embedded in FASTA Input card
Citation DOI	Placeholder	10.5281/zenodo.20681412

3 Installation and Tool Acquisition

3.1 R and Shiny Packages

R version 4.2 or later is required. Install from <https://cran.r-project.org>. The following packages must be installed before launching the app:

```
install.packages(c(
  'shiny', 'bs4Dash', 'DT', 'data.table', 'openxlsx',
  'httr2', 'stringr', 'dplyr', 'shinycssloaders',
  'shinyjs', 'htmltools'
))
```

```
# For mzIdentML / pepXML parsing in Tab 6:
```

```
install.packages('XML')
```

```
# Optional Bioconductor packages:
```

```
if (!requireNamespace('BiocManager', quietly = TRUE))
```

```
  install.packages('BiocManager')
```

```
BiocManager::install(c('Biostrings', 'mzR'))
```

3.2 MS Amanda 3.0 (Primary Search Engine)

MS Amanda 3.0 is a free, standalone peptide identification engine developed at the Institute of Molecular Pathology (IMP), Vienna. It is designed specifically for high-resolution Orbitrap data and requires no Java runtime or commercial licence.

Download: <https://github.com/hgb-bin-proteomics/MSAmanda/releases>

Platform	Binary name	Notes
Windows 10/11 (x64)	MSAmanda.exe	Standalone .exe; no installer needed
Linux (x86_64)	MSAmanda	Requires .NET 6 runtime (see below)
macOS (Intel/Apple Si)	MSAmanda	Requires .NET 6 runtime (see below)

Linux / macOS prerequisite — .NET 6 runtime:

```
# Ubuntu / Debian
```

```
sudo apt-get install -y dotnet-runtime-6.0
```

```
# Fedora / RHEL
```

```
sudo dnf install dotnet-runtime-6.0
```

```
# macOS (Homebrew)
```

```
brew install --cask dotnet-runtime
```

```
# Or download directly:
# https://dotnet.microsoft.com/en-us/download/dotnet/6.0
```

Installation steps:

- Download the zip for your platform from the GitHub releases page.
- Extract to a directory, e.g. `~/MSAmanda/` (Linux/Mac) or `C:\MSAmanda\` (Windows).
- On Linux/Mac: `chmod +x ~/MSAmanda/MSAmanda`
- Optionally set the environment variable (add to `~/.bashrc` or `~/.zshrc`): `export MSAMANDA_EXE=~/MSAmanda/MSAmanda`
- Verify: `MSAmanda --version` should print the version string.

Accepted input formats: mzML, MGF. Output format used by this app: mzIdentML (flag `-f 2`).

3.3 Tide / Crux 4.x (Fallback Search Engine)

Tide is a fast database search engine bundled inside the Crux toolkit, developed at the University of Washington. It is released under the Apache 2.0 licence and has no external dependencies.

Download: <https://crux.ms/download.html>

Platform	Binary name	Notes
Windows 10/11 (x64)	<code>crux.exe</code>	Standalone .exe in the zip
Linux (x86_64)	<code>crux</code>	Statically linked; no dependencies
macOS (Intel/Apple Si)	<code>crux</code>	Universal binary available

Installation steps:

- Download the pre-built binary zip from <https://crux.ms/download.html>.
- Extract to a directory, e.g. `~/crux/` (Linux/Mac) or `C:\crux\` (Windows).
- On Linux/Mac: `chmod +x ~/crux/bin/crux`
- Optionally set the environment variable: `export CRUX_EXE=~/crux/bin/crux`
- Verify: `crux --version` should print the version string.

Accepted input formats: mzML, mzXML, MGF. Output format used by this app: pepXML (`--pepxml-output T`).

3.4 ProteoWizard MSConvert (Raw File Conversion)

Vendor raw files (Thermo `.raw`, Bruker `.d`, Waters `.raw`) must be converted to mzML or MGF before upload. MSConvert from the ProteoWizard suite is the standard tool for this conversion and is free to download.

Download: <https://proteowizard.sourceforge.io>

Recommended command:

```
msconvert input.raw --mzML --filter "peakPicking true 1-"
```

The `peakPicking` filter centroid-picks the spectra, which is required for both MS Amanda and Tide. For Bruker `.d` folders, replace `input.raw` with the folder path.

3.5 Anthropic API Key (AI Assistant)

Tab 7 requires a free or paid Anthropic API key. Obtain one from <https://console.anthropic.com>. The key can be entered in the AI Assistant tab and saved permanently to your `.Renviron` file with the **Save to .Renviron** button, so it is loaded automatically on every subsequent R session.

3.6 Background Proteome Database

The uniqueness filter in Tab 3 requires pre-built `.rds` files for each species. Generate them once by running the bundled script from the app root directory:

```
source('build_background_db.R')
```

This downloads reviewed Swiss-Prot proteomes for human, cynomolgus monkey, and rat from UniProt (approximately 10 minutes, requires internet access). The resulting files (`data/bg_human.rds`, `data/bg_monkey.rds`, `data/bg_rat.rds`) are reused on every subsequent launch. If you already have these files from v0.7, copy them into the v0.8 `data/` folder — no rebuild is needed.

3.7 Summary Table

Component	Required for	Where to get	Free?
R $\geq$ 4.2	App runtime	cran.r-project.org	Yes
R packages (Section 3.1)	App runtime	CRAN / Bioconductor	Yes
MS Amanda 3.0	Tab 6 search (primary)	github.com/hgb-bin-proteomics/MSAmanda	Yes
.NET 6 runtime	MS Amanda on Linux/Mac	dotnet.microsoft.com/download/dotnet/6.0	Yes
Crux 4.x (Tide)	Tab 6 search (fallback)	crux.ms/download.html	Yes
ProteoWizard MSConvert	Raw file conversion	proteowizard.sourceforge.io	Yes
Anthropic API key	Tab 7 AI Assistant	console.anthropic.com	Free tier available
Internet access	<code>build_background_db.R</code> (once)	—	—

4 Application Overview

ADC Peptide Mapper is a seven-tab R Shiny application. Each tab builds on the previous: the digest in Tab 1 feeds the coverage map in Tab 3, the transition list in Tab 4, the heavy labelling in Tab 5, the MS/MS search in Tab 6, and the AI interpretation in Tab 7.

Tab	Name	Purpose
1	Input and Setup	Upload ADC FASTA, select enzyme, missed cleavages, background species, run digest
2	Modifications	Fixed, variable, special, linker biotransformation, and custom PTMs; ADCDB payloads; DAR settings
3	Peptide Results	Browse and filter peptide table; sequence coverage map; co-species uniqueness check
4	Transition List	Select instrument platform and DAR level; generate and download MRM/DIA transition lists
5	Heavy Labelling	Generate SIL-IS light/heavy peptide pairs for quantitative LC-MS/MS
6	MS/MS Search	Run MS Amanda 3.0 or Tide/Crux locally; cross-reference PSMs; FDR estimation
7	AI Assistant	Claude-powered chat for ADC peptide mapping questions

5 Tab 1 — Input and Setup

5.1 ADC FASTA Input

The left panel accepts the ADC sequence in FASTA format via a tabset panel:

- **Upload File** — accepts `.fasta`, `.fa`, or `.txt` files. Multi-chain FASTA is supported; each `>header` entry is treated as a separate chain.

- **Paste Sequence** — the demo Trastuzumab HC + LC sequence is pre-loaded and can be replaced by pasting any FASTA text.

A chain preview table appears below the input, showing the header, length (in amino acids), and the first 60 residues of each chain.

5.2 ADC Name

The ADC Name field (e.g. Trastuzumab-DXd) is used in all output filenames and as the `ADCName` column in transition list exports.

5.3 Enzyme and Cleavage Parameters

Enzyme ID	Cleavage rule
Trypsin	After K or R, not before P
Lys-C	After K
Lys-N	Before K
Asp-N	Before D
Glu-C (E)	After E
Glu-C (E/D)	After E or D
Chymotrypsin	After F, Y, W, L
Arg-C	After R
CNBr	After M
Trypsin + Lys-C	After K or R (not before P) and after K

Missed cleavages (0, 1, or 2) and peptide length filter (fixed at 6–30 AA) are applied during digest. An optional second enzyme enables sequential dual-enzyme digestion.

5.4 Background Proteome

The background proteome is used to flag peptides as unique to the ADC (not present in the host proteome). Three pre-built backgrounds are available: Human, Cynomolgus Monkey, and Rat. A custom FASTA upload option is also provided. The missed cleavage level for the background lookup can be set independently of the digest level.

5.5 Run Digest

The **Run Digest & Uniqueness Check** button is located at the bottom of the FASTA Input card. Clicking it executes the full pipeline: FASTA parsing, enzymatic cleavage, modification application, uniqueness flagging, and navigation to Tab 3. A notification reports the total peptide count and the number unique to the ADC.

6 Tab 2 — Modifications

6.1 Fixed Modifications

Carbamidomethylation (CAM, +57.021464 Da) is always applied to all cysteine residues. This reflects standard iodoacetamide alkylation in sample preparation and cannot be disabled.

6.2 Variable Modifications

Variable modifications generate additional peptide variants. Each selected modification is applied to all eligible residues in the peptide sequence:

- Oxidation (M) +15.994915 Da — methionine sulfoxide
- Propionamide (C) +71.037114 Da — acrylamide adduct
- N-Ethylmaleimide (C) +125.047679 Da — NEM adduct
- Drug-Linker Payload — mass shift applied to C; selected from the ADCDB dropdown or entered as a custom value

6.3 Linker Biotransformation Modifications (new in v0.8)

Five variable modifications model common in-vivo and in-vitro chemical transformations of ADC linker-payload species. These are applied to conjugated cysteine residues:

Biotransformation	Mass shift (Da)	Residue	Reference
Maleimide ring hydrolysis	+18.011	C	Shen 2012, Nat Biotechnol
Succinimide ring-opening	+18.011	C	Alley 2010, Curr Opin Chem Biol
Thioether to sulfoxide oxidation	+15.995	C	Lyon 2015, Nat Biotechnol
Disulfide loss	-31.990	C	Tumey 2014, Bioconjug Chem
Deamidation at conjugation site	+0.984	N	Harris 2001, J Chromatogr B

6.4 Special Modifications

Predefined modifications for common PTMs:

- Deamidation (N) +0.984016 Da
- Deamidation (Q) +0.984016 Da
- Pyroglutamate N-term Q -17.026549 Da
- Pyroglutamate N-term E -18.010565 Da

- Acetylation (N-term) +42.010565 Da
- Phosphorylation (S/T/Y) +79.966331 Da

6.5 Custom Modification Builder

Any modification can be added via the Custom Modification Builder: specify the target residue (single letter), modification name, mass shift (Da), and location (any residue, N-terminus only, or C-terminus only). Custom modifications appear in the peptide table and are carried through to transition lists and heavy labelling.

6.6 DAR Distribution Settings (new in v0.8)

The DAR panel (lower section of Tab 2) configures Drug-to-Antibody Ratio distribution modelling. Select the conjugation chemistry (cysteine thiol-maleimide, lysine NHS-ester / hydrazone, or site-specific) and the maximum DAR level. The app generates a full MRM transition list for each DAR species (DAR0 through DARn) when building the transition list in Tab 4.

7 Tab 3 — Peptide Results

7.1 Value Boxes

Four value boxes at the top of Tab 3 summarise the digest: total peptide variants, peptides unique to the ADC, modified variants, and number of chains detected.

7.2 Peptide Results Table

The main table shows all peptide variants with the following columns: Chain, Peptide Sequence, Modified Sequence, Start, End, Length, Modifications, Mass (Da), and Unique to ADC. Unique peptides are highlighted with a navy badge; non-unique peptides with a grey badge.

7.3 Filters

Three dropdown filters are available above the table: Filter by Chain, Uniqueness (All / Unique to ADC only / Non-unique only), and Modifications (All / Modified only / Unmodified only). Per-column filter rows are also available in the table header.

7.4 Cross-Species Co-Uniqueness Check (new in v0.8)

A dedicated panel allows filtering peptides that are unique across multiple species simultaneously. Select two or more background proteomes (Human, Cynomolgus Monkey, Rat) and click **Run Co-uniqueness Check**. Only peptides absent from all selected backgrounds are retained. This is recommended for multi-species PK/TK studies.

7.5 Excel Export

The **Download Excel Summary** button exports the filtered peptide table as an `.xlsx` file with the ADC name and date in the filename.

7.6 Sequence Coverage Map

A collapsible Sequence Coverage Map card appears below the peptide results table immediately after the digest completes. No MS/MS search is required. The map shows all theoretical peptides from the digest mapped onto each chain sequence using a greedy lane-assignment algorithm that prevents overlapping bars.

What the map shows:

- X-axis: residue position (1 to chain length)
- Y-axis: peptide lanes (stacked rows, greedy assignment by start position)
- Each peptide is a horizontal rectangle spanning its Start to End positions
- Coverage percentage is annotated per chain (e.g. HC: 74.0% covered (333 / 450 AA))

Controls:

Control	Options	Description
Show chain	All chains / individual chain names	Filter the map to one chain or show all chains stacked
Colour by	Uniqueness / Missed cleavages / Peptide length	Uniqueness: navy = unique, grey = non-unique. MC: dark to light blue. Length: viridis.
Show peptide labels	Checkbox (off by default)	Overlay sequence text on peptides ≥ 8 AA wide
Download PNG	Button	Export the current map as a 300 dpi PNG file

Interpretation notes:

- Gaps in coverage indicate peptides outside the 6–30 AA length filter, or not generated by the selected enzyme.
- Non-unique peptides (grey) are present in the background proteome and will not be flagged as ADC-specific in quantitative assays.
- The coverage percentage reflects theoretical in-silico coverage only. Actual MS/MS detection coverage is shown in Tab 6 after a search.
- Increasing missed cleavages (Tab 1) will generally increase coverage by generating longer peptides that span gaps.

8 Tab 4 — Transition List

8.1 Instrument Platform

Select the target instrument from the dropdown. The collision energy (CE) formula and CSV column layout are adjusted per platform:

Platform	CE Formula	Notes
Skyline (generic)	Sciex-style default	Direct Skyline import
Thermo (Orbitrap/TSQ)	Linear, charge-dependent	HCD/CID optimised
SCIEX (QTRAP/TripleTOF)	Empirical, charge-dependent	MRM and SWATH
Bruker (timsTOF)	TIMS-adjusted	PASEF compatible
Agilent (QQQ/QTOF)	Agilent empirical	MRM optimised
Waters (Xevo/Synapt)	Waters empirical	MRM optimised

8.2 DAR Level Selection (new in v0.8)

When DAR modelling is enabled in Tab 2, a DAR level selector appears in Tab 4. Select one or more DAR levels (DAR0 through DARn) to include in the exported transition list. Each DAR level generates separate precursor m/z values reflecting the additional drug-linker payload mass on conjugated peptides.

8.3 Ion Parameters

Fragment ions are calculated as full b and y series: b2 through b(n-1) and y2 through y(n-1), where n is the peptide length. All product ions are singly charged. Precursor charge states 2+, 3+, and 4+ are generated for each peptide.

8.4 Averagine Isotope Envelope (new in v0.8)

An averagine-based isotope distribution (Senko 1995) is computed for each precursor. The monoisotopic peak and the first three isotope peaks (M, M+1, M+2, M+3) are reported in the Excel export to assist with targeted MS method development.

8.5 Filters and Options

- **Unique peptides only** — restrict the transition list to peptides flagged as unique to the ADC in Tab 3.
- **Top N ions per precursor** — limit the number of fragment ions per precursor (0 = all ions).

8.6 Downloads

- **Download Instrument CSV** — platform-specific format for the selected instrument.
 - **Skyline CSV (All Peptides)** — Skyline-compatible CSV with all peptides.
 - **Skyline CSV (Unique Only)** — Skyline-compatible CSV with unique peptides only.
 - **Excel Summary (All Sheets)** — multi-sheet Excel workbook with peptides, transitions, and isotope data.
-

9 Tab 5 — Heavy Labelling

9.1 Isotope Label Presets

Six stable isotope label (SIL) presets are available for generating light/heavy peptide pairs for quantitative LC-MS/MS assays:

Label	Residue	Mass Shift (Da)
13C6 15N2 Lys	K	+8.014199
13C6 15N4 Arg	R	+10.008269
D4 Lys	K	+4.025107
D6 Leu	L	+6.031817
13C6 Leu	L	+6.020129
13C9 15N1 Tyr	Y	+10.009369
Custom	User-defined	User-defined

9.2 Charge States

Select one or more precursor charge states (2+, 3+, 4+) for the heavy peptide list. The app calculates light and heavy m/z values for each peptide x charge combination.

9.3 Output

The Heavy Labelled Peptide Preview table shows: Chain, Sequence, Modified Sequence, Heavy Label, Label Residue, Number of Label Sites, Mass Shift (Da), Charge, Light m/z, Heavy m/z, and Delta m/z.

The Light/Heavy Pair Summary tab shows one row per peptide x charge, useful for verifying isotope incorporation and setting up SRM/PRM assays.

9.4 Downloads

- **Download Heavy Peptides CSV** — full table as CSV.

- **Download Heavy Peptides Excel** — two-sheet workbook: Heavy Peptides and Light/Heavy Pairs.
-

10 Tab 6 — MS/MS Search

Tab 6 runs MS Amanda 3.0 (primary) or Tide/Crux (fallback) on your local machine and cross-references the PSM results against the theoretical peptide list from Tab 3. The digest must be run in Tab 1 before using Tab 6.

10.1 Engine Status Badge

The status badge at the top of the left panel shows which engine is detected and ready. If neither engine is found, the badge shows links to the MS Amanda GitHub releases page and the Crux download page. See Section 3 for installation instructions.

10.2 Engine Executable Path

Leave the executable path field blank for auto-detection. The app searches in the following order: (1) the path entered in the field, (2) the `MSAMANDA_EXE` or `CRUX_EXE` environment variable, (3) common directories (`~/tools/`, `~/bin/`, `~/MSAmanda/`, `~/crux/bin/`), (4) the system `PATH`.

10.3 Spectral File Upload

Upload one or more spectral files or pre-computed result files. The format is auto-detected from the file extension:

Format	Extension	Notes
mzML	.mzML	Recommended open standard; convert from raw with MSConvert
mzXML	.mzXML	Older open standard; supported
MGF	.mgf	Peak list format; supported
mzIdentML	.mzid / .mzidentml	MS Amanda output; upload skips search step
pepXML	.pepxml / .pep.xml	Tide/Crux output; upload skips search step
psm.tsv	psm.tsv	FragPipe / MSFragger legacy; upload skips search step

Format	Extension	Notes
Amanda CSV	.csv	MS Amanda summary CSV; upload skips search step

10.4 Score Filters

The score slider label and range change automatically based on the detected engine. Filters are applied post-search and do not require re-running the search:

- **MS Amanda:** Minimum Amanda Score (0–2000, default 100). Higher scores indicate better matches. Typical threshold: 100–200.
- **Tide/Crux:** Minimum XCorr (0–10, default 1.5). Higher XCorr indicates better matches. Typical threshold: 1.5–2.5.
- **E-value / p-value:** Maximum E-value (default 0.01). Lower values are more stringent.

10.5 Target-Decoy FDR Estimation (new in v0.8)

A target-decoy FDR estimate (Kall 2008) is computed after each search. Decoy sequences are generated by reversing each target peptide. The estimated FDR at the selected score threshold is displayed in the results panel. For bioanalytical method development, an FDR of 1% or lower is recommended.

10.6 MS Amanda Settings

When MS Amanda is used, the app automatically generates a `settings.xml` file from the Tab 1 and Tab 2 parameters. The following defaults are applied:

Parameter	Default	Notes
Precursor tolerance	20 ppm	Suitable for Orbitrap data
Fragment tolerance	0.02 Da	Suitable for high-resolution MS2
Instrument type	HighRes	Optimised for Orbitrap
Fixed mod	Carbamidomethyl on C (+57.021464 Da)	Always applied
Variable mod	Oxidation on M (+15.994915 Da)	Always included
Variable mod 2	Drug-linker payload on C	Only if payload selected in Tab 2
Enzyme	From Tab 1 setting	Mapped to MS Amanda enzyme names
Missed cleavages	From Tab 1 setting	0, 1, or 2

Parameter	Default	Notes
Peptide length	6–30 AA	Matches Tab 1 digest filter
Output format	mzIdentML (-f 2)	Parsed automatically by the app

10.7 Results

After a successful search, three sub-tabs are shown in the right panel:

- **Sequence Coverage** — per-chain coverage bar showing detected peptides (navy = unique to ADC, grey = non-unique) overlaid on the full chain sequence. Coverage percentage is annotated per chain.
- **Detected vs Theoretical** — table of all theoretical peptides with detection status, PSM count, best score, and best E-value.
- **Modification Summary** — horizontal bar chart of PSM counts per detected modification type.

10.8 Downloads

- **Download PSM Table (CSV)** — filtered PSM table.
- **Download Coverage Summary (CSV)** — per-chain coverage statistics.

11 Tab 7 — AI Assistant (new in v0.8)

The AI Assistant tab embeds an Anthropic Claude chat interface pre-configured with ADC and proteomics context. It can help interpret peptide mapping results, explain mass spectrometry concepts, suggest experimental designs, and troubleshoot workflows.

11.1 Setup

1. Go to the **AI Assistant** tab.
2. Paste your Anthropic API key into the key field.
3. Click **Save to .Renviron** — the key is written to your `~/.Renviron` file and loaded automatically on every future R session.
4. Select a Claude model from the dropdown (populated dynamically from your account's available models via the `/v1/models` endpoint).
5. Type your question and click **Send** (or press Enter).

11.2 Context Toggle

Enable **Include digest context** to automatically attach the current ADC name, number of chains detected, enzyme, missed cleavages setting, and total peptide count to each message sent to the API. This provides the model with relevant experimental parameters without requiring manual copy-paste.

11.3 API Key Security

The API key is stored only in your local `~/.Renv` file and is never transmitted anywhere other than the Anthropic API endpoint (`api.anthropic.com`). When deploying to `shinyapps.io`, set the key as a **shinyapps.io Environment Variable** (app Settings → Environment Variables → `ANTHROPIC_API_KEY`) — never hard-code it in `app.R` or commit it to version control.

12 Output Reference

12.1 Peptide Table Columns

Column	Description
Chain	FASTA header of the chain this peptide belongs to
Peptide Sequence	Unmodified amino acid sequence (uppercase)
Modified Sequence	Sequence with modification annotations, e.g. <code>PEPTC[CAM]IDE</code>
Start	1-based start position in the chain
End	1-based end position in the chain
Length	Peptide length in amino acids
Modifications	Semicolon-separated list of applied modifications with positions
Mass (Da)	Monoisotopic mass of the modified peptide
Unique to ADC	Yes (navy) if not present in the background proteome; No (grey) otherwise

12.2 Transition List Columns

Column	Description
ADCName	ADC name from Tab 1
ProteinName	Chain identifier (from FASTA header)
Chain	Full chain header

Column	Description
PeptideSequence	Unmodified peptide sequence
ModifiedSequence	Peptide sequence with modification annotations
PrecursorCharge	Precursor charge state (2, 3, or 4)
PrecursorMz	Precursor m/z (monoisotopic, 6 decimal places)
FragmentIon	Fragment ion label, e.g. b5 or y12
ProductMz	Product ion m/z (monoisotopic, 6 decimal places)
CollisionEnergy	Calculated CE in eV (platform-specific formula)
UniqueToADC	TRUE if peptide is unique to the ADC
DARLevel	DAR species (DAR0, DAR1, ..., DARn); present when DAR modelling enabled

13 MS/MS Search Engine Setup

This section provides step-by-step instructions for installing MS Amanda 3.0 and Tide/Crux on each supported platform. For download URLs and a summary table, see Section 3.

13.1 MS Amanda 3.0 — Windows

- Download `MSAmanda_Windows.zip` from the GitHub releases page.
- Extract to `C:\MSAmanda\`.
- In Tab 6, paste `C:\MSAmanda\MSAmanda.exe` into the executable path field, or set `MSAMANDA_EXE=C:\MSAmanda\MSAmanda.exe` in your `.Renviron` file.
- Verify: open a Command Prompt and run `MSAmanda.exe --version`.

13.2 MS Amanda 3.0 — Linux

```
# Install .NET 6 runtime
```

```
sudo apt-get install -y dotnet-runtime-6.0
```

```
# Download and extract
```

```
mkdir ~/MSAmanda && cd ~/MSAmanda
```

```
wget https://github.com/hgb-bin-proteomics/MSAmanda/releases/latest/download/MSAmanda_Linux.zip
```

```
unzip MSAmanda_Linux.zip
```

```
chmod +x MSAmanda
```

```
# Add to environment (add to ~/.bashrc)
```

```
export MSAMANDA_EXE=~/.MSAmanda/MSAmanda
```

```
# Verify
```

```
~/.MSAmanda/MSAmanda --version
```

13.3 MS Amanda 3.0 — macOS

```
# Install .NET 6 runtime via Homebrew
```

```
brew install --cask dotnet-runtime
```

```
# Download and extract
```

```
mkdir ~/.MSAmanda && cd ~/.MSAmanda
```

```
# Download MSAmanda_macOS.zip from GitHub releases
```

```
unzip MSAmanda_macOS.zip
```

```
chmod +x MSAmanda
```

```
# Add to environment (add to ~/.zshrc)
```

```
export MSAMANDA_EXE=~/.MSAmanda/MSAmanda
```

```
# Verify
```

```
~/.MSAmanda/MSAmanda --version
```

13.4 Tide / Crux — All Platforms

```
# Download from https://crux.ms/download.html
```

```
# Extract and make executable (Linux/Mac)
```

```
mkdir ~/crux && cd ~/crux
```

```
# Extract the downloaded zip
```

```
chmod +x bin/crux
```

```
# Add to environment
```

```
export CRUX_EXE=~/.crux/bin/crux
```

```
# Verify
```

```
~/crux/bin/crux --version
```

13.5 Setting Environment Variables in R

The most reliable way to set MSAMANDA_EXE or CRUX_EXE (or ANTHROPIC_API_KEY) is via the .Renviron file, which is read by R on startup:

```
# Open .Renviron in R:
```

```

usethis::edit_r_environ()

# Add one or more of these lines:
MSAMANDA_EXE=/home/user/MSAmanda/MSAmanda
CRUX_EXE=/home/user/crux/bin/crux
ANTHROPIC_API_KEY=sk-ant-...

# Save the file and restart R.

```

14 Deployment to shinyapps.io

To make the app available on the web via shinyapps.io:

```

# One-time account setup
install.packages('rsconnect')
rsconnect::setAccountInfo(
  name    = 'your-account-name',    # from shinyapps.io -> Account -> Profile
  token   = 'YOUR_TOKEN',          # from shinyapps.io -> Account -> Tokens
  secret  = 'YOUR_SECRET'
)

# Deploy (from the project root directory)
source('deploy.R')

```

After deploying, go to your app page on shinyapps.io, open **Settings** → **Environment Variables**, and add:

```
ANTHROPIC_API_KEY = sk-ant-...
```

This powers the AI Assistant tab. Never hard-code the key in `app.R` or commit it to version control.

The background `.rds` files in `data/` are large (50–150 MB each). The shinyapps.io free tier has a 1 GB bundle limit. Edit `INCLUDE_SPECIES` in `deploy.R` if you need to reduce the bundle size.

15 Running Unit Tests and Mass Accuracy Benchmark

```

setwd('path/to/ADC_Peptide_Mapper_v0.8')
source('R/digest.R')
source('R/modifications.R')
source('R/transitions.R')

```

```

source('R/isotopes.R')
source('R/dar.R')
source('tests/test_masses.R') # 79 tests
source('tests/benchmark_mass_accuracy.R') # 10 reference peptides

```

All 79 unit tests should pass with a summary of **79 / 79 PASSED**. The mass accuracy benchmark verifies sub-mDa accuracy against four IgG1 tryptic reference peptides (GPSVFPLAPSSR, ELASGLSFPVGFK, CASIQKFGR, DTLMISR) and six b/y ions from CASIQKFGR vs. Mascot Ion Calculator reference values. All 10 benchmarks should pass with errors of ≤ 0.05 mDa.

16 Scientific Background

16.1 Monoisotopic Masses

All mass calculations use monoisotopic masses (most abundant isotope of each element). Residue masses are standard NIST values. The proton mass used for m/z calculation is 1.007276 Da.

16.2 Peptide Mass Calculation

The neutral monoisotopic mass of a peptide is calculated as the sum of residue masses plus water (18.010565 Da), plus the sum of all applied modification mass shifts. The precursor m/z for charge state z is:

$$m/z = \frac{M_{\text{neutral}} + z \times 1.007276}{z}$$

16.3 Fragment Ion Calculation

b-ions are N-terminal fragments; y-ions are C-terminal fragments. For a peptide of length n :

- b-ions: b2 through b($n-1$). Mass = sum of N-terminal residue masses + modifications.
- y-ions: y2 through y($n-1$). Mass = sum of C-terminal residue masses + modifications + H₂O.

All product ions are singly charged ($z = 1$). No a-ions are generated.

16.4 Averagine Isotope Envelope

The isotope distribution for each peptide is estimated using the averagine model (Senko 1995), which approximates a peptide of given mass as a scaled version of the average amino acid composition. The model predicts the relative abundances of the monoisotopic peak (M) and the first three isotope peaks (M+1, M+2, M+3) without requiring knowledge of the exact elemental composition.

16.5 Uniqueness Checking

A peptide is flagged as unique to the ADC if its unmodified sequence is not present in the background proteome at the selected missed cleavage level. The background proteomes are sourced from UniProt Swiss-Prot reviewed entries and digested in silico with trypsin at MC 0, 1, and 2.

16.6 Target-Decoy FDR Estimation

False discovery rate (FDR) is estimated using the target-decoy approach (Kall 2008). Decoy sequences are generated by reversing each target peptide. For a score threshold t , the FDR is estimated as:

$$\text{FDR}(t) = \frac{\text{decoy PSMs}(t)}{\text{target PSMs}(t)}$$

16.7 MS Amanda Scoring

MS Amanda uses a probabilistic scoring model based on the hypergeometric distribution to evaluate the likelihood that a peptide-spectrum match (PSM) occurred by chance. Higher Amanda scores indicate better matches. The score is not directly comparable to Hyperscore (MSFragger) or XCorr (Tide).

16.8 Tide XCorr Scoring

Tide uses the cross-correlation score (XCorr) from the original SEQUEST algorithm. XCorr measures the similarity between the observed spectrum and the theoretical spectrum for a candidate peptide. Higher XCorr values indicate better matches.

17 Troubleshooting

17.1 App does not launch

- Confirm all required R packages are installed (Section 3.1).
- Check that `data/bg_human.rds` exists; run `source('build_background_db.R')` if not.
- Run `shiny::runApp('app.R')` from the project root directory.

17.2 Background database build fails

- Confirm internet access is available.
- The cRAP download uses FTP (`ftp://ftp.thegpm.org/fasta/cRAP/crap.fasta`); some corporate firewalls block FTP. If FTP is blocked, the build script falls through to a bundled minimal cRAP set (6 sequences).

- If UniProt download fails, retry after a few minutes (rate limiting).

17.3 Digest error: “the condition has length > 1”

- This error indicates a version mismatch between the app and the R source files. Ensure all files in R/ are from the same v0.8 release zip.

17.4 AI Assistant returns API error

- Verify the API key is correct (starts with `sk-ant-`).
- Check that the key has not expired or been revoked at <https://console.anthropic.com>.
- Confirm internet access from within R: `httr2::request('https://api.anthropic.com')`
|> `httr2::req_perform()`

17.5 MS Amanda not detected

- Confirm the binary is executable: `file ~/MSAmanda/MSAmanda` should show an ELF or Mach-O binary, not a zip.
- Set `MSAMANDA_EXE` in `.Renviron` and restart R (Section 12.5).
- On macOS, .NET 6 runtime must be installed before MS Amanda will run.

17.6 Mass accuracy is off by ~1 Da

- This usually indicates a monoisotopic vs. average mass confusion. Confirm that the input sequence uses standard single-letter codes and that no non-standard residues are present.

17.7 shinyapps.io deployment bundle too large

- Set `INCLUDE_SPECIES` in `deploy.R` to exclude species you do not need (e.g. `c('human')` only).
- Ensure the `tests/` directory and large local files are excluded via `EXCLUDE_PATTERNS` in `deploy.R`.

18 Citation

If you use ADC Peptide Mapper in your research, please cite:

Wase, N. (2026). *ADC Peptide Mapper* (Version 0.8) [Software]. <https://doi.org/10.5281/zenodo.20681412>

If you use the MS/MS Search tab (Tab 6) with MS Amanda, also cite:

Dorfer V, Pichler P, Stranzl T, Stadlmann J, Taus T, Winkler S, Mechtler K.
MS Amanda, a Universal Identification Algorithm Optimized for High Accu-

racy Tandem Mass Spectra. *J Proteome Res.* 2014;13(8):3679–3684. <https://doi.org/10.1021/pr500202e>

If using Tide/Crux, also cite:

McIlwain S, Tamura K, Kertesz-Farkas A, Grant CE, Diamant B, Frewen B, Howbert JJ, Hoopmann MR, Kall L, Eng JK, MacCoss MJ, Noble WS. Crux: Rapid Open Source Protein Tandem Mass Spectrometry Analysis. *J Proteome Res.* 2014;13(10):4488–4491. <https://doi.org/10.1021/pr500741y>

For linker biotransformation references and all other citations, see `CITATION.cff`.

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For research use only. Monoisotopic masses throughout. Background databases sourced from UniProt Swiss-Prot reviewed proteomes (Human, Cynomolgus Monkey, Rat).